

Seohee Choy

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EDUCATION

University of Wisconsin - Madison

Bachelor of Science in Computer Science and Data Science | GPA: 3.86 / 4.0

Madison, WI

September 2023 - May 2027

EXPERIENCE

Robust Intelligence and Robotics Lab @ KAIST | [Lab Website](#)

Daejeon, South Korea

Undergraduate Researcher, Visiting Scholar (Advisor: Professor DaeHyung Park)

November 2025 - Present

- Conduct research on active multi-view visual inputs to improve manipulation robustness under occlusion.
- Design and manufacture remote gimbal camera modules using **Fusion 360, Raspberry Pi, BLDC motors, and FOC drivers** to improve visual access and reduce task-space occlusions in a **UR5e** workspace.
- Implement **Human-in-the-Loop SERL (HIL-SERL)** policies in **Isaac Sim** to incorporate human guidance into autonomous policy learning for sequential manipulation tasks such as pick-and-place.

Vision Lab @ LG AI Research | [LG AI Talk Concert](#)

Seoul, South Korea

AI Engineer Intern

May 2025 - October 2025

- Developed automated pipelines to curate large-scale multimodal QA datasets across **DocVQA, ChartQA, MMMU, AI2D, and K-DTCBench**. Used **CLIP embeddings** to retrieve failure-related samples for **EXAONE-VL**, LG AI Research's **vision-language model (VLM)**.
- Converted public and **300K+** confidential datasets into VLM-compatible training formats with strict validation of image presence, modality completeness, and size constraints for OCR, charts, tables, captions, and QAs.
- Extended an internal **TypeScript, Node.js, FastAPI, PostgreSQL** data labeling platform with new annotation features to support human-in-the-loop Korean benchmark construction.

University of Wisconsin - Madison, Department of Computer Science

Madison, WI

Peer Mentor for CS 540 (Introduction to AI)

August 2024 - December 2024

- Mentored students on **Python-based AI/ML projects** covering search, neural networks, and computer vision.
- Provided 6 hours of weekly 1:1 support to **10+ students** by translating complex algorithmic concepts into debuggable implementation strategies.

PROJECTS

FIRST Tech Challenge Korea (Branksome Hall Asia Robotics Academy)

December 2020 - May 2023

- Led a **25 member team** across mechanical engineering, software programming, and logistics.
- Developed **Java-based vision recognition and autonomous navigation logic** via live webcam input.

FarmBot (Smart Farming System) | [Media](#)

August 2020 - April 2023

- Co-led the hardware assembly, deployment, and sensor calibration of the **CNC-based FarmBot system**.
- Developed control logic to automate planting, watering, and weeding based on real-time sensor feedback.

Full-Stack Study Room Reservation System | [Demo](#)

May 2022 - May 2023

- Built a **TypeScript, React, and Firebase** reservation platform for **400+** students, implementing real-time validation, duplicate booking prevention, and synchronized reservation management.

AWARDS

Dean's List, University of Wisconsin-Madison

Fall 2023, Spring 2024

The Wisconsin Alumni Association Korea Chapter (WAAK) Scholarship

Fall 2023

USA Computing Olympiad (USACO) Silver & Bronze Division

March 2021

ACSL All-Star Round Award

March 2021

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, SQL, HTML/CSS

AI/ML & Frameworks: PyTorch, TensorFlow, Isaac Sim, OpenCV, React, React Native, FastAPI, Node.js

Tools & Platforms: Linux, Git, Fusion 360, Docker, Google Cloud Platform (GCP), Firebase